

PLATFORM TECHNICAL BRIEF · V1.0

PMG Platform

Proactive Value Chain Resilience

An analytical platform that traces your supply chain to the raw material source — eight levels deep — scores what it finds at both the supplier and material level, and delivers prescriptive guidance grounded in engineering-grade intelligence.

VERSION	SCOPE	CLASSIFICATION
v1.0 — Proactive Resilience	Technical evaluation	Confidential

WHY EXISTING TOOLS ARE STRUCTURALLY INSUFFICIENT

Risk lives where your tools stop looking

Every supply chain risk methodology in production today operates on supplier-visible data. The foundational academic work in supply chain risk quantification — the Ford-MIT Time-to-Recover model, Interfaces 45(5), 2015 — explicitly acknowledged that sub-tier data collection was deferred by design due to data availability constraints. That constraint was reasonable in 2015. The risks it leaves invisible are not acceptable today.

The material constraints that drive the longest disruptions — rare earth processing concentration, specialty alloy tooling lock-in, chemical synthesis bottlenecks — live at Level 4 through Level 7 of the bill of materials. A supplier's declared Time-to-Recover reflects their own operational capacity. It does not include the physics of the material chain beneath them.

T1 DECLARED TTR

8-12 wk

Supplier's own estimate
of recovery time



PMG COMPUTED TTR

52+ wk

Computed from material chain
beneath the supplier

The gap originates at Level 5: a sintered rare earth magnet production process with maximum manufacturing complexity, concentrated in a handful of facilities globally. The Tier 1 supplier's estimate is honest. It does not include what they cannot see beneath themselves. PMG was designed to close that gap.

PLATFORM AT A GLANCE

170K+

Nodes across 8 BOM
levels — live modelled
European BEV dataset

540+

Suppliers mapped T1-T8
in the same modelled
dataset

Dual

Supplier AND material
nodes scored
simultaneously

100%

XAI audit trail coverage
across all scores

Scale figures are from a representative European BEV archetype running live on the PMG platform. Customer deployments are scoped to the agreed plant and vehicle line — depth scales with engagement scope.

HOW THE PLATFORM IS BUILT

A material graph, not a supplier map

PMG models every bill of materials as an eight-level directed graph. Each node is either a supplier entity or a material specification. Both are scored independently.

REPRESENTATIVE EPS SUPPLY CHAIN — NDFEB PATH



PMG Analysis · European BEV Archetype · Illustrative

Dual-Node Scoring

Both supplier and material nodes are scored independently in the same graph pass. A low-risk supplier can carry catastrophic material-level exposure three levels below them. Both scores are required.

4-Vector Risk Model

Manufacturing · Financial · Operational · Geopolitical. Each vector is scored per node. The weighted sum produces the node’s overall risk score. Every weight is traceable through the XAI chain.

TTR Computed from Material Physics

Time-to-Recover is not self-reported. It is computed from the graph — tracing the actual constraint: tooling recut timelines, chemical re-qualification periods, geographic concentration at the ore source.

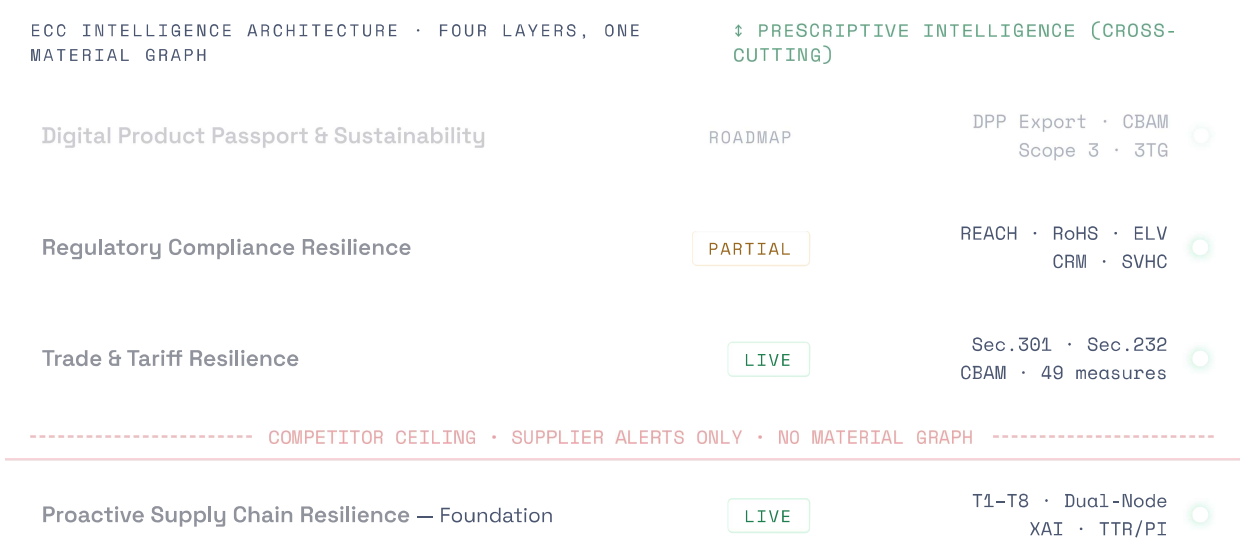
XAI Audit Trail — 100% Coverage

Every score carries a structured reasoning chain traceable to the computation that produced it, at the moment it was produced. Auditable by compliance teams. Ingestible by AI copilot systems.

PLATFORM ARCHITECTURE

Four intelligence layers on one material graph

Each layer is independently deployable. All four draw from the same PMG material graph. A Prescriptive Intelligence spine runs as a cross-cutting orchestration layer across all domains.



Prescriptive Intelligence — an agentic orchestration layer extended by partner intelligence networks — runs as a vertical cross-cut across all four domains. Available in each layer context, not as a separate module. See Section 6.

WHAT YOUR TEAMS CAN DO WITH IT

Six analytical surfaces. Six questions already on your leadership's desk.

Each surface is a standalone capability with its own data model, scoring logic, and explainability chain — all grounded in the same material graph.

#	SURFACE	WHAT IT ANSWERS	KEY CAPABILITIES
01	Risk patterns in your material structure	Where are your structural single-points-of-failure — the deep-tier nodes holding multiple product lines hostage?	Diamond detection · Concentration clusters · Single-source fan-outs · Tooling lock-in distributions · Portfolio-level pattern signatures
02	Geographic concentration	How many strategic suppliers share the same country — and where does material-level concentration compound that exposure?	Supplier-level concentration view · Country-material heat map · Assembly exposure by geo cluster · Jurisdiction-level aggregate risk
03	Tariff and trade exposure	What does a tariff schedule change actually cost — by material, by assembly, with alternate sourcing options?	HS code attribution at material nodes · Cost exposure by assembly · Automatic impact propagation · Alternate sourcing simulation
04	Deep-tier supplier financial health	Which T1 suppliers are financially distressed — and which sub-tier suppliers, invisible to your current monitoring, are deteriorating right now?	T1-T8 supplier mapping · PMG-prioritised sub-tier monitoring · 6-dimension health scoring · Distress phase trajectory · Assembly exposure linkage
05	Regulatory and compliance exposure	Which materials in your assemblies are approaching restriction — before the restriction takes effect?	CAS-level mapping · ELV / REACH SVHC / RoHS · CRM scarcity · Restriction timeline tracking · Auto-propagation on regulation updates
06	Risk quadrant positioning	Where in the risk landscape does each supplier and material actually sit — and what action does each quadrant require?	4-quadrant scatter · Strategic vs hidden risk · Portfolio prioritisation view · Graph-computed position (not self-reported)

BEYOND RISK IDENTIFICATION

From finding alternatives to understanding trade-offs

Standard prescriptive tools recommend an alternative supplier. ECC evaluates whether the alternative is viable — materially, structurally, and regulatorily — before the recommendation surfaces. The output brings procurement and engineering to the same table.

PATHWAY 01

Material substitution evaluation

When a material is at risk, the prescriptive layer queries a database of 570,000+ materials across 80+ international standards for global equivalents. Mechanical and physical properties of each candidate are compared against the performance specification. Compliance status against ELV, REACH, and RoHS is validated. Multi-criteria scoring ranks candidates by functional fit, supply concentration, and sustainability profile. Output: an engineering-grade substitution brief for procurement and engineering jointly.

PATHWAY 02

Sourcing viability simulation

Once a candidate material is identified, the prescriptive layer maps qualified manufacturers across a live supply graph of dependencies up to ten tiers deep — country-of-origin, capacity indicators, dependency strengths. "Does an alternative exist?" is answered alongside "Is this executable, by whom, and what does their dependency profile look like?" Simulation runs before the recommendation lands with the team.

PATHWAY 03

Cross-domain consequence analysis

A prescriptive action that reduces tariff exposure might change the DPP compliance baseline or introduce a new geographic concentration. Because the orchestration layer draws from the same material graph across all three domains — proactive, tariff, regulatory — every recommendation is evaluated for cross-domain consequences before it surfaces. The procurement lead and compliance lead see the same picture.

RESILIENCE FABRIC

Beyond the computable. The prescriptive layer also draws from a curated knowledge base built from hundreds of supply chain research papers and documented historical incidents — containing the mitigation strategies that were deployed, what worked, and what did not. When the platform identifies a critically important sub-tier supplier under financial stress, it surfaces options beyond replacement: supply chain finance instruments, inventory pre-build programmes, collaborative recovery frameworks. The right action is not always the obvious one.

Prescriptive intelligence is a cross-cutting layer available across all four intelligence domains. Partner-extended capabilities — material substitution intelligence and live supply graph simulation — are available through the ECC

partner network. Partners available on request.

HOW PMG CONNECTS TO YOUR ENVIRONMENT

Your stack stays intact. Your data stays in your infrastructure.

Integration model

PMG integrates with SAP and other SCM systems through published APIs or SAP's own agentic connectors. Existing procurement workflows, ERP configuration, and supplier relationships are untouched.

- **Read-only access.** PMG does not write to your systems.
- **API-native.** No direct database access. SAP ecosystem compatible.
- **Need-to-know scope.** Integration covers only the data relevant to the scoped engagement.
- **Auditable.** Every data access through the integration layer is logged.
- **Open signal ingestion.** Architecture supports customer-owned assessment data as an additional input signal — scored alongside platform data without requiring code changes.

Deployment model

ECC deploys into your own Azure subscription and your own Neo4j instance. We provision and configure. You own the infrastructure.

- **Azure deployment in your tenancy.** No shared cloud infrastructure operated by ECC.
- **Neo4j in your account.** The material graph runs in an instance you own and control.
- **Data sovereignty.** Your supply chain data never moves to ECC's environment.
- **Isolated per customer.** No multi-tenant deployment model.

SECURITY

Primary security mechanism is architectural: customer-owned infrastructure means supply chain data never moves to a shared environment. ECC does not hold, process, or store customer data outside the customer's own cloud accounts.

Security architecture documentation, threat model, network isolation design, data access controls, and testing results are available on request and prepared to enterprise standard. The ECC founding team includes a Microsoft-certified Cybersecurity and Solutions Architect.

DPP DUAL RETURN

PMG builds its material graph using science-quality life cycle inventory data — CAS-level compound data, international LCI datasets, critical minerals sourcing intelligence. This is the same data

foundation required for EU Digital Product Passport compliance. The data work ECC does for risk intelligence simultaneously builds the customer's DPP compliance baseline. One data investment. Two strategic returns.

GETTING STARTED

A validated output in 4–6 weeks. No permanent commitment.

Scoped to one plant and one vehicle line. Every finding is traceable. Every number has a computation behind it.

PHASE 1

Weeks 1–2

Data Connection

API integration with your BOM and supplier data. Read-only access, defined scope. No raw data extraction, no persistent access beyond the integration layer. Scope agreed and documented before connection.

PHASE 2

Weeks 2–4

Graph Build & Scoring

PMG builds the material genealogy graph for the scoped vehicle line. Every node scored. Every TTR computed from the material chain. Every reasoning chain generated and stored. Deep-tier supplier mapping completed and prioritised.

PHASE 3

Weeks 5–6

Validated Output Review

Findings walkthrough with your procurement and risk team. Every finding traceable. Every number has a computation behind it. You decide whether and how to extend coverage. No automatic commitments.

About this document — v1.0

This brief covers PMG's Proactive Value Chain Resilience segment. ECC's platform also includes a Reactive Resilience segment — an agentic copilot system for supply chain response workflows, event-triggered analysis, and guided recovery decision-making. Reactive Resilience documentation is available on request and will be incorporated into v2 of this brief.

Ready to see what PMG finds in your supply chain?

One plant. One vehicle line. 4–6 weeks. No stack changes required.

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